

Exact block reductions in Fourier-modal scattering theory for layered periodic media

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Abstract

The Fourier modal method represents Maxwell's equations in a periodic layer by a finite set of n reciprocal-lattice harmonics. In a generic formulation, the modal problem has dimension $2n$ and the stable composition of layers by a Redheffer product requires dense cubic algebra. We derive exact reductions of these operators for structures occurring frequently in diffraction theory. For an isotropic homogeneous layer, a permutation to the per-harmonic basis $(E_{x,h}, E_{y,h})$ decomposes the Maxwell operator into independent 2×2 blocks at arbitrary incidence. Consequently, an arbitrary homogeneous multilayer is propagated in $O(nN_u)$ operations, where N_u is the number of homogeneous interfaces. For a y -invariant patterned layer, the modal matrix is block diagonal at normal incidence and block lower triangular under conical illumination. Its spectrum is therefore obtained from two $n \times n$ non-Hermitian eigenproblems; in the conical case we give an explicit reconstruction of the coupled eigenvectors. We also derive a block contraction for quadratic volume observables that avoids the $4n \times 4n$ intermediate matrices of the direct expression. Numerical experiments against grcwa and S4 verify the scattering observables and Fourier fields to the accuracy of the reference discretization. For an extreme- ultraviolet absorber on a 40-period Mo/Si mirror, the homogeneous-stack reduction accelerates the scattering stage by approximately 3.4–3.7, while the combined one-dimensional reductions give an order-of-magnitude or greater acceleration for line gratings. The analysis separates exact operator reductions from approximation-dependent convergence acceleration and identifies the patterned-layer eigensystem as the remaining asymptotic bottleneck.

Keywords: Fourier modal method; rigorous coupled-wave analysis; diffraction gratings; scattering matrix; conical incidence; block-triangular eigenproblem.

1 Introduction

Rigorous coupled-wave analysis (RCWA), also called the Fourier modal method (FMM), is the workhorse for computing the optical response of layered, laterally periodic photonic structures: diffraction gratings, photonic crystals, thin-film solar cells, metasurfaces, and the phase-shifting absorber stacks of extreme-ultraviolet (EUV) lithography [1–3]. The method expands the electromagnetic field in a truncated set of n planewave (Fourier) harmonics in the plane of periodicity, reduces Maxwell's equations in each laterally structured layer to a $2n \times 2n$ non-Hermitian eigenproblem, and propagates the modal amplitudes across the stack with a scattering-matrix (Redheffer) recursion [4, 5]. Reflectivity, transmissivity, per-order diffraction efficiencies, near-field intensities, and absorbed-power distributions all follow from the assembled S -matrix and the per-layer modal data.

The numerical difficulty is controlled by the number n of retained spatial harmonics. A patterned layer requires a non-Hermitian eigenproblem of order $2n$, while stable composition across interfaces requires dense matrix products and linear solves of the same order. Both stages scale cubically and store $O(n^2)$ complex numbers. Large-period gratings make this cost especially severe: at wavelength λ , a period L_x

admits on the order of $2L_x/\lambda$ propagating diffraction channels before evanescent harmonics are included. Repeated evaluations in wavelength, angle, or geometric parameter studies compound the cost.

A generic implementation spends its time in two algebraic operations:

1. $O((2n)^3)$ for the (generally non-Hermitian, complex) eigendecomposition (13) of each laterally structured layer, and
2. $O((2n)^3)$ dense matrix products and inversions *per layer interface* in the Redheffer recursion (17).

As a representative geometry, consider vacuum | absorber | cap | $40\times(\text{Mo/Si})$ | substrate. The single patterned layer is followed by approximately 80 homogeneous interfaces. Treating those interfaces as generic dense scatterers repeats cubic work even though translational invariance forbids coupling between distinct in-plane wavevectors in every homogeneous layer. The purpose of this paper is to expose that invariance, and related invariant subspaces of the patterned operator, directly in the finite-dimensional Maxwell system.

1.1 Context and main results

The Fourier-modal method was introduced for diffraction gratings by Moharam and Gaylord [1] and refined into a stable, efficient formulation for binary gratings in [2]. The stability of the layer-recursion step was settled by Li’s comparison of the R -, T -, and S -matrix algorithms [5], establishing the scattering-matrix (Redheffer) formulation as the method of choice for thick and lossy stacks; our work uses exactly this recursion [6]. The convergence of the modal expansion for discontinuous media was analyzed by Li [7], whose inverse rule (and its generalization, the normal-vector field / fast Fourier factorization [8]) removes the algebraic convergence bottleneck of the naive Laurent rule for high-contrast metallic gratings. The scattering-matrix treatment of patterned multilayer photonic structures was systematized by Whittaker and Culshaw [4].

Convergence acceleration by Fourier factorization and normal-vector fields reduces the harmonic order needed at a specified accuracy [7, 8]. The present work addresses a different question: given a fixed Fourier discretization, which parts of its algebra are exactly reducible? The principal results are:

1. A homogeneous isotropic layer is a direct sum of n two-dimensional polarization spaces, for any reciprocal-lattice truncation and any Bloch wavevector. A homogeneous stack therefore admits an exact per-harmonic scattering recursion.
2. For a y -invariant isotropic profile, the patterned modal matrix splits into two n -dimensional spectral problems. At normal incidence the matrix is block diagonal; at conical incidence it is block lower triangular, and the off-diagonal eigenvector block is obtained from a Sylvester equation.
3. Quadratic volume observables can be contracted using $2n$ -dimensional blocks, without constructing the direct $4n$ -dimensional expression.

These are identities of the discretized Maxwell operator. They introduce no material, weak-coupling, or paraxial approximation and are complementary to factorization rules that improve convergence with n .

The open-source solvers `grcwa` [9] and `S4` [3] are used as independent numerical references. The derivation is developed in sections 2 to 4; implementation and numerical tests are reported in sections 8 and 9.

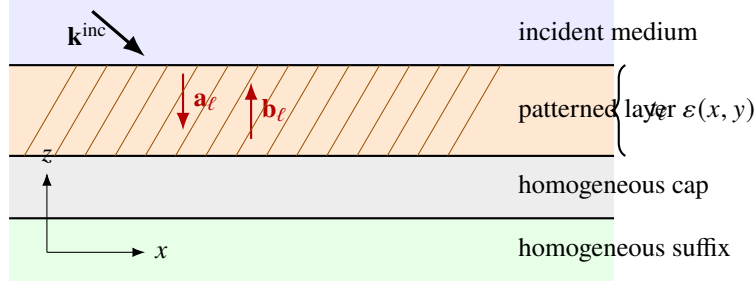


Figure 1: Layered periodic medium and modal-amplitude convention. Only the patterned layers couple distinct reciprocal-lattice harmonics. Every homogeneous region retains polarization coupling within a fixed harmonic but does not couple different harmonics.

2 Fourier-modal formulation

2.1 Geometry, harmonics, and notation

Consider a stack of layers, each invariant along z (the propagation axis), with the in-plane permittivity $\varepsilon(x, y)$ periodic on a lattice with direct vectors $\mathbf{L}_1, \mathbf{L}_2$ and reciprocal vectors $\mathbf{K}_1, \mathbf{K}_2$. The field is expanded in the n retained harmonics $\{\mathbf{G}_i\}_{i=1}^n \subset \{m_1\mathbf{K}_1 + m_2\mathbf{K}_2\}$, so that every in-plane quantity is represented by n Fourier coefficients and the transverse wavevectors of harmonic i are

$$k_{x,i} = k_{x0} + (m_1\mathbf{K}_1 + m_2\mathbf{K}_2) \cdot \hat{x}, \quad k_{y,i} = k_{y0} + (m_1\mathbf{K}_1 + m_2\mathbf{K}_2) \cdot \hat{y}, \quad (1)$$

with k_{x0}, k_{y0} fixed by the incidence angles. We write $\mathbf{k}_x, \mathbf{k}_y \in \mathbb{C}^n$ for the diagonal entries $k_{x,i}, k_{y,i}$, and let $\mathbf{q} \in \mathbb{C}^{2n}$ denote the (unknown) vertical propagation constants. The total block size is $N = 2n$; all scattering matrices are $N \times N$.

Figure 1 fixes the geometry and the amplitude convention used throughout the paper. Within layer ℓ , $\mathbf{a}_\ell$ denotes modes whose phase advances in the positive z direction and $\mathbf{b}_\ell$ denotes modes whose phase advances in the negative z direction. For evanescent waves the branch convention introduced below turns these into decaying solutions in the corresponding half space.

For clarity, two orderings of the $2n$ transverse components will be used. The *component-major* ordering is

$$\mathbf{u}_c = [u_{x,1}, \dots, u_{x,n}, u_{y,1}, \dots, u_{y,n}]^\top, \quad (2)$$

whereas the *harmonic-major* ordering is

$$\mathbf{u}_h = [u_{x,1}, u_{y,1}, u_{x,2}, u_{y,2}, \dots, u_{x,n}, u_{y,n}]^\top. \quad (3)$$

They are related by a permutation matrix Π , with $\mathbf{u}_h = \Pi\mathbf{u}_c$ and $\Pi^{-1} = \Pi^\top$. No approximation is associated with this change of basis; it merely makes different sparsity patterns visible.

2.2 The layer eigenproblem

We derive the eigenproblem from Maxwell's equations. With the $e^{+i\omega t}$ time convention, the curl equations in a source-free, non-magnetic medium are

$$\nabla \times \mathbf{E} = -i\omega\mathbf{H}, \quad \nabla \times \mathbf{H} = i\omega\varepsilon\mathbf{E}. \quad (4)$$

For a Bloch wave with in-plane Bloch vector $\mathbf{k}_\parallel^{(0)}$, a field component f has the truncated expansion

$$f(x, y, z) = \sum_{j=1}^n f_j(z) \exp\left[i(\mathbf{k}_\parallel^{(0)} + 2\pi\mathbf{G}_j) \cdot \mathbf{r}_\parallel\right]. \quad (5)$$

Orthogonality of these plane waves turns ∂_x and ∂_y into multiplication by iK_x and iK_y . Multiplication by a periodic permittivity is not diagonal: if $\varepsilon(\mathbf{r}_{\parallel}) = \sum_{\mathbf{g}} \hat{\varepsilon}_{\mathbf{g}} e^{2\pi i \mathbf{g} \cdot \mathbf{r}_{\parallel}}$, then

$$(\varepsilon f)_i = \sum_{j=1}^n \hat{\varepsilon}_{\mathbf{G}_i - \mathbf{G}_j} f_j, \quad C_{ij} = \hat{\varepsilon}_{\mathbf{G}_i - \mathbf{G}_j}. \quad (6)$$

This convolution is the sole source of coupling between distinct harmonics in a layer.

Fourier-expanding the transverse field components $e_x, e_y, h_x, h_y \in \mathbb{C}^n$ and stacking $\mathbf{e}_t = [e_x; e_y]$, $\mathbf{h}_t = [h_x; h_y]$ (in the ordering $[E_{x,1} \dots E_{x,n}; E_{y,1} \dots E_{y,n}]$), the in-plane derivatives ∂_x, ∂_y become multiplications by the diagonal matrices $K_x = \text{diag}(k_{x,i})$, $K_y = \text{diag}(k_{y,i})$, and the permittivity becomes the convolution operator C . Eliminating the longitudinal components from (4),

$$e_z = -\frac{1}{\omega} C^{-1} (K_y h_x - K_x h_y), \quad h_z = \frac{1}{\omega} (K_y e_x - K_x e_y), \quad (7)$$

To see the first relation explicitly, take the z component of $\nabla \times \mathbf{H} = i\omega \varepsilon \mathbf{E}$:

$$iK_x h_y - iK_y h_x = i\omega C e_z, \quad (8)$$

and solve for e_z ; the z component of the other curl equation gives h_z . Substitution of these two longitudinal fields into the four remaining curl components yields the first-order system

$$\frac{d}{dz} \begin{bmatrix} e_x \\ e_y \\ h_x \\ h_y \end{bmatrix} = \mathbf{A} \begin{bmatrix} e_x \\ e_y \\ h_x \\ h_y \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{0} \end{bmatrix}, \quad (9)$$

with the $2n \times 2n$ blocks

$$\mathbf{A}_{12} = \begin{bmatrix} -\frac{i}{\omega} K_x C^{-1} K_y & -i\omega I + \frac{i}{\omega} K_x C^{-1} K_x \\ i\omega I - \frac{i}{\omega} K_y C^{-1} K_y & \frac{i}{\omega} K_y C^{-1} K_x \end{bmatrix}, \quad (10)$$

$$\mathbf{A}_{21} = \begin{bmatrix} \frac{i}{\omega} K_x K_y & i\omega C - \frac{i}{\omega} K_x^2 \\ -i\omega C + \frac{i}{\omega} K_y^2 & -\frac{i}{\omega} K_y K_x \end{bmatrix}. \quad (11)$$

Because $\mathbf{A}_{12}\mathbf{A}_{21}$ and $\mathbf{A}_{21}\mathbf{A}_{12}$ share their eigenvalues, the eigenvalues of $\mathbf{A}$ occur in $\pm$ -pairs, $\{\pm i q_j\}$, and eliminating the magnetic components gives the second-order equation

$$\frac{d^2 \mathbf{e}_t}{dz^2} = -\mathbf{A}_{12} \mathbf{A}_{21} \mathbf{e}_t, \quad (12)$$

whose eigenvalues are $-q_j^2$. In the planewave (Fourier-modal) basis the same problem is written as the $2n \times 2n$ non-Hermitian eigenproblem

$$\mathbf{M} \Phi = \Phi \text{diag}(\mathbf{q}^2), \quad (13)$$

whose eigenvalues are the squared vertical propagation constants q_j^2 , and the modal fields vary as $e^{\pm i q_j z}$. The matrix $\mathbf{M}$ is built from the permittivity convolution (Toeplitz) matrix $C_{ij} = \hat{\varepsilon}(\mathbf{G}_i - \mathbf{G}_j)$,

$$\mathbf{M} = \text{ep2} K - \mathbf{k} \mathbf{k}^T, \quad \text{ep2} = \text{blkdiag}(C, C), \quad (14)$$

with $E = C^{-1}$, and

$$K = \omega^2 \mathbf{I} - (\mathbf{J}_k E) \mathbf{J}_k^T, \quad \mathbf{J}_k = \begin{bmatrix} -\text{diag}(\mathbf{k}_y) \\ \text{diag}(\mathbf{k}_x) \end{bmatrix}, \quad \mathbf{k} = \begin{bmatrix} \text{diag}(\mathbf{k}_x) \\ \text{diag}(\mathbf{k}_y) \end{bmatrix}. \quad (15)$$

The branch cut $\mathbf{q}_i \mapsto -\mathbf{q}_i$ when $\text{Im } \mathbf{q}_i < 0$ selects the physically decaying (evanescent) modes in $+z$. Here ω is the (possibly complex) angular frequency; the lossy case $\text{Im } \omega \neq 0$ or complex ε is included throughout. For a homogeneous layer of permittivity ε_0 , $E = 1/\varepsilon_0$ and the same formulas apply with C replaced by the scalar ε_0 . (The placement of E in (15) is consistent with the inverse-factorization treatment of the discontinuous normal field component [7]; alternative Fourier-factorization rules may be substituted without changing the block arguments below.)

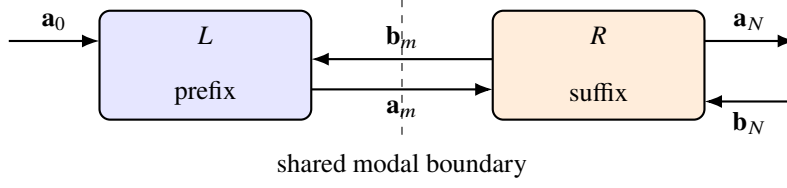


Figure 2: Redheffer composition as elimination of the two amplitudes at the shared internal boundary. The same diagram applies when the right subnetwork is represented compactly by independent 2×2 blocks.

2.3 S-matrix (Redheffer) recursion

Two adjacent layers $l, l+1$ are connected by the interface matrices

$$\mathbf{Q}_l = \Phi_l^{-1} \Phi_{l+1}, \quad \mathbf{P}_l = \text{diag}(\mathbf{q}_l) (\mathbf{K}_l \Phi_l)^{-1} \mathbf{K}_{l+1} \Phi_{l+1} \text{diag}(1/\mathbf{q}_{l+1}), \quad (16)$$

and the transmission matrices $\mathbf{T}_{11} = \frac{1}{2}(\mathbf{Q} + \mathbf{P})$, $\mathbf{T}_{12} = \frac{1}{2}(\mathbf{Q} - \mathbf{P})$. The scattering matrix of the composite stack follows from the Redheffer star product [5, 6]: with diagonal phase factors $\mathbf{d}_\ell = \text{diag}(e^{iq_\ell t_\ell})$ for layer thickness t_ℓ , one update of the accumulated S -matrix is

$$\begin{aligned} \mathbf{P}_1 &= (\mathbf{T}_{11} - \mathbf{d}_1 \mathbf{S}_{12} \mathbf{T}_{12})^{-1}, \\ \mathbf{S}_{11} &\leftarrow \mathbf{P}_1 \mathbf{d}_1 \mathbf{S}_{11}, \quad \mathbf{S}_{12} \leftarrow \mathbf{P}_1 (\mathbf{d}_1 \mathbf{S}_{12} \mathbf{T}_{11} - \mathbf{T}_{12}) \mathbf{d}_2, \\ \mathbf{S}_{21} &\leftarrow \mathbf{S}_{21} + \mathbf{S}_{22} \mathbf{T}_{12} \mathbf{S}_{11}, \quad \mathbf{S}_{22} \leftarrow \mathbf{S}_{22} (\mathbf{T}_{11} \mathbf{d}_2 + \mathbf{T}_{12} \mathbf{S}_{12}). \end{aligned} \quad (17)$$

Each update costs a handful of dense $N \times N$ matrix products plus one LU -based solve.

The reason for using a scattering matrix rather than a transfer matrix is stability. A transfer matrix carries both exponentially growing and decaying solutions through a thick layer. Even when the physical answer is bounded, the intermediate transfer matrix can overflow or lose all significant digits. The scattering matrix instead relates incoming amplitudes to outgoing amplitudes and keeps the two directions separated. In the convention used in (17),

$$\begin{bmatrix} \mathbf{a}_{\ell+1} \\ \mathbf{b}_\ell \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}_\ell \begin{bmatrix} \mathbf{a}_\ell \\ \mathbf{b}_{\ell+1} \end{bmatrix}. \quad (18)$$

To derive the update, introduce the internal amplitudes $\mathbf{a}_m, \mathbf{b}_m$ at the shared interface of two subnetworks L and R :

$$\mathbf{a}_m = L_{11} \mathbf{a}_0 + L_{12} \mathbf{b}_m, \quad (19)$$

$$\mathbf{b}_m = R_{21} \mathbf{a}_m + R_{22} \mathbf{b}_N. \quad (20)$$

Elimination gives

$$(I - L_{12} R_{21}) \mathbf{a}_m = L_{11} \mathbf{a}_0 + L_{12} R_{22} \mathbf{b}_N. \quad (21)$$

Equation (27) is obtained by solving (21) and substituting the result into the outgoing relations. Thus the Redheffer product is not a heuristic matrix operation: it is Gaussian elimination of the amplitudes at an internal interface.

3 Structure of homogeneous layers

3.1 A per-harmonic block lemma

Lemma 1. *Let the $2n$ field components be ordered by harmonic as $[E_{x,1}, E_{y,1}, E_{x,2}, E_{y,2}, \dots, E_{x,n}, E_{y,n}]$ (the per-harmonic interleaving). Then for a homogeneous isotropic layer of permittivity ϵ_0 ,*

$$\mathbf{K} = \text{blkdiag}(\mathbf{K}^{(1)}, \dots, \mathbf{K}^{(n)}), \quad \mathbf{K}^{(h)} = \omega^2 \mathbf{I}_2 - \frac{1}{\epsilon_0} \begin{bmatrix} k_{y,h}^2 & -k_{x,h} k_{y,h} \\ -k_{x,h} k_{y,h} & k_{x,h}^2 \end{bmatrix}, \quad (22)$$

the propagation constants satisfy $\mathbf{q}_h = \mathbf{q}_{n+h} = q_h$ with $q_h^2 = \varepsilon_0 \omega^2 - k_{x,h}^2 - k_{y,h}^2$, and $\Phi = \mathbf{I}_{2n}$.

Proof. We first work in the component-major ordering, where the proof is a direct computation. From (15) with $E = 1/\varepsilon_0$ and $\mathbf{J}_k = [-\text{diag}(\mathbf{k}_y); \text{diag}(\mathbf{k}_x)]$, the product

$$\mathbf{J}_k \mathbf{J}_k^\top = \begin{bmatrix} -\text{diag}(\mathbf{k}_y) \\ \text{diag}(\mathbf{k}_x) \end{bmatrix} \begin{bmatrix} -\text{diag}(\mathbf{k}_y) & \text{diag}(\mathbf{k}_x) \end{bmatrix} = \begin{bmatrix} \text{diag}(\mathbf{k}_y^2) & -\text{diag}(\mathbf{k}_x \mathbf{k}_y) \\ -\text{diag}(\mathbf{k}_x \mathbf{k}_y) & \text{diag}(\mathbf{k}_x^2) \end{bmatrix} \quad (23)$$

is a 2×2 block matrix whose blocks are themselves *diagonal*. In the standard $[E_{x,1} \dots E_{x,n}; E_{y,1} \dots E_{y,n}]$ ordering, each harmonic h therefore couples only to itself: the (h, h) entry of the $(1, 1)$ -block, the $(h, n+h)$ entry of the $(1, 2)$ -block, etc. Forming the 2×2 submatrix of rows/columns $\{h, n+h\}$ for each h and applying the interleaving permutation that orders the components as $[E_{x,1}, E_{y,1}, E_{x,2}, E_{y,2}, \dots]$ yields exactly (22); different harmonics are never coupled.

It remains to verify the eigensystem. Because (22) is block diagonal, each harmonic obeys an independent dispersion relation. Writing the modal fields as $e^{i(q_h z)} e^{i(k_{x,h} x + k_{y,h} y)}$ and inserting into Maxwell's equations (4) with $\varepsilon = \varepsilon_0$ gives $k_{x,h}^2 + k_{y,h}^2 + q_h^2 = \varepsilon_0 \omega^2$, i.e. $q_h^2 = \varepsilon_0 \omega^2 - k_{x,h}^2 - k_{y,h}^2$. The corresponding eigenvectors are the coordinate vectors, so in the interleaved ordering $\Phi = \mathbf{I}_{2n}$ and the propagation constants $\mathbf{q} = [\mathbf{q}_x; \mathbf{q}_y]$ satisfy $\mathbf{q}_x = \mathbf{q}_y = q_h$ componentwise. $\square$

The permutation argument is central to the reduction. In the component-major basis the four $n \times n$ blocks are diagonal, but the two entries belonging to one diffraction order are separated in memory. Let Π denote the permutation from component-major to harmonic-major ordering. Then

$$\Pi K \Pi^\top = \text{blkdiag}(K^{(1)}, \dots, K^{(n)}). \quad (24)$$

Equation (24) is an identity, not an approximation. It shows directly why a homogeneous layer can be applied or inverted by n independent 2×2 operations. A dense implementation hides this fact by using the component-major ordering and consequently performs arithmetic on known zero entries.

The lemma is exact for *any* harmonic set — rectangular, circular, or anisotropic (n_x, n_y) — and for *any* incidence, including oblique ($k_{y0} \neq 0$): the only inputs are the per-harmonic pairs $(k_{x,h}, k_{y,h})$. It is the reason the quasi-1D “diagonal core” tricks known in the literature for $k_y = 0$ generalize directly to full 2D.

Corollary 2. *The interface matrices (16) between two homogeneous layers, and therefore the scattering matrix of any all-homogeneous stack, are 2×2 -block-diagonal in the interleaved ordering.*

Proof. For two homogeneous layers $\Phi_\ell = \Phi_{\ell+1} = \mathbf{I}_{2n}$, so the first interface matrix in (16) is $\mathbf{Q}_\ell = \mathbf{I}_{2n}$. The second is $\mathbf{P}_\ell = \text{diag}(\mathbf{q}_\ell) K_\ell^{-1} K_{\ell+1} \text{diag}(1/\mathbf{q}_{\ell+1})$. By Lemma 1 each of the four factors is block-diagonal with 2×2 blocks (the diag factors are diagonal, hence block-diagonal), and products and inverses of block-diagonal matrices are block-diagonal. Per harmonic h ,

$$\mathbf{P}^{(h)} = \frac{q_{\ell,h}}{q_{\ell+1,h}} (K_\ell^{(h)})^{-1} K_{\ell+1}^{(h)}, \quad (25)$$

with $K^{(h)}$ as in (22). The transmission matrices $\mathbf{T}_{11} = \frac{1}{2}(\mathbf{I} + \mathbf{P})$, $\mathbf{T}_{12} = \frac{1}{2}(\mathbf{I} - \mathbf{P})$, the diagonal phase factors $\mathbf{d}_\ell$, and every matrix in the star update (17) are then block-diagonal as well: the inverse $(\mathbf{T}_{11} - \mathbf{d}_1 \mathbf{S}_{12} \mathbf{T}_{12})^{-1}$ of a block-diagonal matrix is block-diagonal, and all products preserve the structure. By induction over the recursion (17), the S -matrix of any all-homogeneous range is 2×2 -block-diagonal. $\square$

Each harmonic's 2×2 recursion is thus fully independent of the others; only the per-harmonic pairs $(k_{x,h}, k_{y,h})$ enter, through (22).

Table 1: Cost of the S -matrix of a stack with a patterned prefix and an N_{layers} -layer uniform suffix ($N = 2n$).

	generic	per-harmonic suffix
uniform interfaces	$O((2n)^3 N_{\text{layers}})$	$O(n N_{\text{layers}})$
patterned prefix	$O((2n)^3)$ per layer	same
assembly	$O((2n)^3)$	$O((2n)^3)$ (once)

3.2 The per-harmonic uniform-suffix algorithm

Consider a stack split at the start m of its trailing all-uniform suffix $[m, L)$, i.e. layers $m, \dots, L - 1$ are homogeneous. Corollary 2 suggests the following exact factorization.

1. *Prefix.* Compute $\mathbf{L} = S(0, m)$ by the general dense star recursion (17) over the (few) layers that contain the pattern, including the boundary homogeneous layer $m - 1$.
2. *Suffix.* Compute the S -matrix of the all-uniform range $[m, L)$ by the per-harmonic 2×2 recursion

$$\begin{aligned} \mathbf{P}^{(h)} &= \frac{q_{\ell, h}}{q_{\ell+1, h}} (\mathbf{K}_{\ell}^{(h)})^{-1} \mathbf{K}_{\ell+1}^{(h)}, \quad \mathbf{T}_{11}^{(h)} = \frac{1}{2}(\mathbf{I}_2 + \mathbf{P}^{(h)}), \quad \mathbf{T}_{12}^{(h)} = \frac{1}{2}(\mathbf{I}_2 - \mathbf{P}^{(h)}), \\ \mathbf{M}^{(h)} &= (\mathbf{T}_{11}^{(h)} - d_{1, h} \mathbf{S}_{12}^{(h)} \mathbf{T}_{12}^{(h)})^{-1}, \quad \mathbf{S}_{11}^{(h)} = d_{1, h} \mathbf{S}_{11}^{(h)} \mathbf{M}^{(h)}, \dots \end{aligned} \quad (26)$$

for each harmonic $h = 1, \dots, n$, storing the four 2×2 blocks per harmonic. The suffix is thereby built in $O(n(L - m))$ operations, each involving a 2×2 inverse.

3. *Assembly.* Join the dense prefix and the block-diagonal suffix by the overlapping cascade (27) below ($\mathbf{L}$ and $\mathbf{R}$ share the boundary layer m).

$$\mathbf{M} = (\mathbf{I} - \mathbf{L}_{12} \mathbf{R}_{21})^{-1}, \quad \mathbf{S}_{11} = \mathbf{R}_{11} \mathbf{M} \mathbf{L}_{11}, \quad \mathbf{S}_{12} = \mathbf{R}_{11} \mathbf{M} \mathbf{L}_{12} \mathbf{R}_{22} + \mathbf{R}_{12}, \quad \dots \quad (27)$$

Derivation of the overlapping cascade. With the scattering convention $[a_{\text{out}}; b_{\text{in}}] = S[a_{\text{in}}; b_{\text{out}}]$, the prefix $\mathbf{L} = S(0, m)$ and the suffix $\mathbf{R} = S(m, L)$ both reference the amplitudes (a_m, b_m) on the shared layer m :

$$a_m = \mathbf{L}_{11} a_0 + \mathbf{L}_{12} b_m, \quad b_m = \mathbf{R}_{21} a_m + \mathbf{R}_{22} b_L, \quad a_L = \mathbf{R}_{11} a_m + \mathbf{R}_{12} b_L. \quad (28)$$

Eliminating a_m, b_m between the first two relations gives $(\mathbf{I} - \mathbf{L}_{12} \mathbf{R}_{21}) a_m = \mathbf{L}_{11} a_0 + \mathbf{L}_{12} \mathbf{R}_{22} b_L$, i.e. $a_m = \mathbf{M}(\mathbf{L}_{11} a_0 + \mathbf{L}_{12} \mathbf{R}_{22} b_L)$ with $\mathbf{M} = (\mathbf{I} - \mathbf{L}_{12} \mathbf{R}_{21})^{-1}$. Substituting into a_L yields $\mathbf{S}_{11} = \mathbf{R}_{11} \mathbf{M} \mathbf{L}_{11}$ and $\mathbf{S}_{12} = \mathbf{R}_{11} \mathbf{M} \mathbf{L}_{12} \mathbf{R}_{22} + \mathbf{R}_{12}$; the backward equations give $\mathbf{S}_{21} = \mathbf{L}_{21} + \mathbf{L}_{22} \mathbf{R}_{21} \mathbf{M} \mathbf{L}_{11}$ and $\mathbf{S}_{22} = \mathbf{L}_{22} \mathbf{R}_{21} \mathbf{M} \mathbf{L}_{12} \mathbf{R}_{22} + \mathbf{L}_{22} \mathbf{R}_{22}$. Because $\mathbf{R}_{21}, \mathbf{R}_{22}$ are 2×2 -block-diagonal, the products $\mathbf{L}_{12} \mathbf{R}_{21}$ and $\mathbf{R}_{11}(\cdot)$ are $O(n)$ block scalings rather than dense products; the only dense $O((2n)^3)$ operations in the whole suffix assembly are the single matrix inverse $(\mathbf{I} - \mathbf{L}_{12} \mathbf{R}_{21})^{-1}$ and the two dense products $\mathbf{M} \mathbf{L}_{11}, \mathbf{M}(\mathbf{L}_{12} \mathbf{R}_{22})$.

Table 1 summarizes the scaling. For a stack whose patterned layers are few (as in the EUV-mask geometry: vacuum | absorber | cap | 40×(Mo/Si) | substrate), the entire multilayer suffix is reduced from 80 dense $O((2n)^3)$ star products to $O(n \cdot 82)$ scalar- 2×2 operations, and the residual dense work is one prefix and one cascade.

3.3 Relation to quasi-1D and periodic-stack methods

When the harmonic set is the x -only row and $k_{y, h} = 0$ for all h , the blocks (22) are diagonal and (26) reduces to independent *scalar* Fresnel-like recursions per harmonic — the classical quasi-1D trick. When the suffix is periodic (a repeated bilayer), the per-harmonic recursion handles it as cheaply as the periodic cores exponentiated by the Redheffer cascade; no periodicity assumption is required. The factorization therefore subsumes both the quasi-1D diagonal core and periodic binary exponentiation as special cases.

4 Exact decompositions of the patterned-layer eigenproblem

The remaining $O((2n)^3)$ bottleneck is the eigenproblem (13) of the patterned layer. We show that for the two most common structures it decouples exactly into two $n \times n$ problems, at $4\times$ fewer eigensolver flops, with a constructive formula for the eigenvectors. Throughout, the harmonic set is the 1D row of a y -invariant grating, so that $\mathbf{k}_y = \kappa_y \mathbf{1}$ is a constant vector, but we emphasize that the block-diagonal case also covers certain 2D patterns (see Section 10).

The following distinction is useful. The homogeneous-layer reduction follows from the absence of lateral material modulation and is valid for every harmonic set. The patterned-layer reduction instead requires a special relation between the material convolution operator and the transverse wavevector. In a y -invariant grating, the retained harmonics have the form $\mathbf{G}_h = (m_h, 0)$, so

$$K_y = \kappa_y I_n, \quad K_x = D_x = \text{diag}(k_{x,1}, \dots, k_{x,n}). \quad (29)$$

The scalar κ_y need not vanish: it is nonzero for conical or oblique illumination. This is precisely why the oblique case is not TE/TM separated, but still has a triangular spectral structure.

4.1 Block-diagonal modal operator at normal incidence

At normal incidence, $k_{x0} = k_{y0} = 0$. For a y -invariant profile one also has $\mathbf{k}_y = \mathbf{0}$. Thus the block in (15) reduces to

$$\mathbf{J}_k = \begin{bmatrix} \mathbf{0} \\ \text{diag}(\mathbf{k}_x) \end{bmatrix}.$$

Writing $D_x = \text{diag}(\mathbf{k}_x)$ gives

$$\mathbf{J}_k \mathbf{E} = \begin{bmatrix} \mathbf{0} \\ D_x \mathbf{E} \end{bmatrix}, \quad (\mathbf{J}_k \mathbf{E}) \mathbf{J}_k^\top = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & D_x \mathbf{E} D_x \end{bmatrix}, \quad (30)$$

so that $\mathbf{K} = \text{diag}(\omega^2 \mathbf{I}, \omega^2 \mathbf{I} - D_x \mathbf{E} D_x)$. Similarly $\mathbf{k} = [D_x; \mathbf{0}]$ gives $\mathbf{k} \mathbf{k}^\top = \text{diag}(D_x^2, \mathbf{0})$, and $\text{ep2} = \text{blkdiag}(C, C)$. Hence

$$\mathbf{M} = \text{ep2} \mathbf{K} - \mathbf{k} \mathbf{k}^\top = \begin{bmatrix} C \omega^2 - D_x^2 & \mathbf{0} \\ \mathbf{0} & C(\omega^2 \mathbf{I} - D_x \mathbf{E} D_x) \end{bmatrix} \equiv \begin{bmatrix} \mathbf{M}_{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_{22} \end{bmatrix}, \quad (31)$$

with, using $C \mathbf{E} = \mathbf{I}$,

$$\mathbf{M}_{11} = \omega^2 C - D_x^2, \quad \mathbf{M}_{22} = \omega^2 C - C D_x C^{-1} D_x. \quad (32)$$

The $2n \times 2n$ eigenproblem (13) therefore decouples exactly into two $n \times n$ problems, $\text{eig}(\mathbf{M}) = \text{eig}(\mathbf{M}_{11}) \cup \text{eig}(\mathbf{M}_{22})$, $\Phi = \text{blkdiag}(\Phi_1, \Phi_2)$, reducing the eigensolver flops by a factor four. (For a D4-symmetric 2D pattern at normal incidence the off-diagonal blocks also vanish exactly, and the same decomposition applies.)

4.2 Block-lower-triangular modal operator at oblique incidence

At oblique incidence $k_{y0} \neq 0$ the 1D harmonic set still has $\mathbf{k}_y = \kappa_y \mathbf{1}$ (a constant vector), so that $D_y = \text{diag}(\mathbf{k}_y) = \kappa_y \mathbf{I}$ and $\mathbf{J}_k = [-\kappa_y \mathbf{I}; D_x]$. Evaluating (15)–(14) blockwise,

$$\mathbf{J}_k \mathbf{E} = \begin{bmatrix} -\kappa_y \mathbf{E} \\ D_x \mathbf{E} \end{bmatrix}, \quad (\mathbf{J}_k \mathbf{E}) \mathbf{J}_k^\top = \begin{bmatrix} \kappa_y^2 \mathbf{E} & -\kappa_y \mathbf{E} D_x \\ -\kappa_y D_x \mathbf{E} & D_x \mathbf{E} D_x \end{bmatrix}, \quad (33)$$

so that

$$\mathbf{K} = \begin{bmatrix} \omega^2 \mathbf{I} - \kappa_y^2 \mathbf{E} & \kappa_y \mathbf{E} D_x \\ \kappa_y D_x \mathbf{E} & \omega^2 \mathbf{I} - D_x \mathbf{E} D_x \end{bmatrix}, \quad \mathbf{k} \mathbf{k}^\top = \begin{bmatrix} D_x^2 & \kappa_y D_x \\ \kappa_y D_x & \kappa_y^2 \mathbf{I} \end{bmatrix}. \quad (34)$$

The four blocks of $\mathbf{M} = \text{ep2 K} - \mathbf{k}\mathbf{k}^\top$ are

$$\mathbf{M}_{11} = C(\omega^2 \mathbf{I} - \kappa_y^2 \mathbf{E}) - D_x^2 = \omega^2 C - \kappa_y^2 \mathbf{I} - D_x^2, \quad (35)$$

$$\mathbf{M}_{12} = C \kappa_y \mathbf{E} D_x - \kappa_y D_x = \kappa_y (C \mathbf{E} - \mathbf{I}) D_x = \mathbf{0}, \quad (36)$$

$$\mathbf{M}_{21} = C \kappa_y D_x \mathbf{E} - \kappa_y D_x = \kappa_y (C D_x C^{-1} - D_x) \neq \mathbf{0}, \quad (37)$$

$$\mathbf{M}_{22} = C(\omega^2 \mathbf{I} - D_x \mathbf{E} D_x) - \kappa_y^2 \mathbf{I} = \omega^2 C - \kappa_y^2 \mathbf{I} - C D_x C^{-1} D_x. \quad (38)$$

Because $\mathbf{M}_{12} = \mathbf{0}$, $\mathbf{M}$ is block-lower-triangular,

$$\mathbf{M} = \begin{bmatrix} \mathbf{M}_{11} & \mathbf{0} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{bmatrix}, \quad (39)$$

and $\det(\mathbf{M} - \lambda \mathbf{I}) = \det(\mathbf{M}_{11} - \lambda \mathbf{I}) \det(\mathbf{M}_{22} - \lambda \mathbf{I})$: the eigenvalues split exactly into two n -dimensional problems $\text{eig}(\mathbf{M}) = \text{eig}(\mathbf{M}_{11}) \cup \text{eig}(\mathbf{M}_{22})$, a $4\times$ reduction in eigensolver flops. The eigenvectors, however, remain coupled through the nonzero lower-left block $\mathbf{M}_{21}$; their exact reconstruction is the subject of the next subsection.

4.3 Exact reconstruction of the coupled eigenvectors

Because $\mathbf{M}$ is block-lower-triangular, the eigenvectors of the two eigenvalue families are constructed as follows.

This construction is an instance of block diagonalization. Suppose $\mathbf{M}_{11}$ and $\mathbf{M}_{22}$ are diagonalizable and their spectra do not intersect. Seek a similarity transformation

$$\mathbf{X} = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{Y} & \mathbf{I} \end{bmatrix} \quad (40)$$

that removes the lower-left block. Direct multiplication gives

$$\mathbf{X}^{-1} \mathbf{M} \mathbf{X} = \begin{bmatrix} \mathbf{M}_{11} & \mathbf{0} \\ \mathbf{M}_{21} + \mathbf{M}_{22} \mathbf{Y} - \mathbf{Y} \mathbf{M}_{11} & \mathbf{M}_{22} \end{bmatrix}. \quad (41)$$

Thus $\mathbf{Y}$ must solve the Sylvester equation

$$\mathbf{M}_{22} \mathbf{Y} - \mathbf{Y} \mathbf{M}_{11} = -\mathbf{M}_{21}. \quad (42)$$

The spectral decompositions $\mathbf{M}_{11} = \Phi_A \Lambda_A \Phi_A^{-1}$ and $\mathbf{M}_{22} = \Phi_C \Lambda_C \Phi_C^{-1}$ convert (42) into

$$\Lambda_C Z - Z \Lambda_A = -\Phi_C^{-1} \mathbf{M}_{21} \Phi_A, \quad Z = \Phi_C^{-1} \mathbf{Y} \Phi_A. \quad (43)$$

Since the two diagonal matrices act by row and column scaling,

$$Z_{ij} = -\frac{(\Phi_C^{-1} \mathbf{M}_{21} \Phi_A)_{ij}}{\lambda_{C,i} - \lambda_{A,j}}. \quad (44)$$

This explains both the reconstruction formula and its failure mode: when an A and C eigenvalue approach one another, the Sylvester operator becomes ill-conditioned and the full $2n$ eigensolve is the appropriate fallback.

Columns for $\lambda_{A,j} = \text{eig}_j(\mathbf{M}_{11})$. Let $\mathbf{M}_{11} = \Phi_A \Lambda_A \Phi_A^{-1}$ with columns $\Phi_A = [\mathbf{a}_1, \dots, \mathbf{a}_n]$ and $\mathbf{M}_{11} \mathbf{a}_j = \lambda_{A,j} \mathbf{a}_j$, and let $\mathbf{M}_{22} = \Phi_C \Lambda_C \Phi_C^{-1}$. Writing an eigenvector as $[\mathbf{a}_j; \mathbf{v}_j]$, the second block row of $(\mathbf{M} - \lambda_{A,j} \mathbf{I})[\mathbf{a}_j; \mathbf{v}_j] = \mathbf{0}$ reads

$$(\mathbf{M}_{22} - \lambda_{A,j} \mathbf{I}) \mathbf{v}_j = -\mathbf{M}_{21} \mathbf{a}_j, \quad (45)$$

the first block row being satisfied identically. When $\lambda_{A,j} \notin \text{eig}(\mathbf{M}_{22})$ the resolvent of $\mathbf{M}_{22}$ gives

$$\mathbf{v}_j = -(\mathbf{M}_{22} - \lambda_{A,j}\mathbf{I})^{-1}\mathbf{M}_{21}\mathbf{a}_j = -\Phi_C \text{diag}\left(\frac{1}{\lambda_{C,i} - \lambda_{A,j}}\right)\Phi_C^{-1}\mathbf{M}_{21}\mathbf{a}_j. \quad (46)$$

Columns for $\lambda_{C,j} = \text{eig}_j(\mathbf{M}_{22})$. Writing an eigenvector as $[\mathbf{u}_j; \mathbf{c}_j]$, the second block row $(\mathbf{M}_{22} - \lambda_{C,j}\mathbf{I})\mathbf{c}_j = 0$ is satisfied by $\mathbf{c}_j = \Phi_C[:, j]$; the first block row $(\mathbf{M}_{11} - \lambda_{C,j}\mathbf{I})\mathbf{u}_j = \mathbf{0}$ then forces $\mathbf{u}_j = \mathbf{0}$ provided $\lambda_{C,j} \notin \text{eig}(\mathbf{M}_{11})$. The corresponding modal columns are therefore $[\mathbf{0}; \Phi_C[:, j]]$.

Collecting both families gives the block-lower-triangular modal matrix

$$\Phi = \begin{bmatrix} \Phi_A & \mathbf{0} \\ \mathbf{V} & \Phi_C \end{bmatrix}, \quad \mathbf{V} = -\Phi_C \left((\Phi_C^{-1}\mathbf{M}_{21}\Phi_A) \oslash \mathbf{D} \right), \quad \mathbf{D}_{ij} = \lambda_{C,i} - \lambda_{A,j}, \quad (47)$$

where $\oslash$ denotes element-wise division. If $\mathbf{X} = \Phi_C^{-1}\mathbf{M}_{21}\Phi_A$, then the (i, j) entry of the Hadamard quotient is

$$(\mathbf{X} \oslash \mathbf{D})_{ij} = \frac{X_{ij}}{\lambda_{C,i} - \lambda_{A,j}},$$

which agrees column by column with (46). The reconstruction (47) costs two $n \times n$ eigendecompositions, one $n \times n$ solve $(\Phi_C^{-1}\mathbf{M}_{21}\Phi_A)$, and two $O(n^3)$ products, versus one $2n \times 2n$ eigendecomposition; the results are numerically identical to the full solve (verified to 17 digits in the scattering observables). If a pair of eigenvalues becomes exactly degenerate ($\min_{ij} |\mathbf{D}_{ij}| \rightarrow 0$) the resolvent (46) is singular and the code falls back to the full $2n \times 2n$ solve; the detection is exact (off-diagonal-block norm test), so the result is never silently wrong.

5 Polarization-decoupled scattering at normal incidence

At normal incidence with block-diagonal $\mathbf{K}, \Phi, \mathbf{q}$ (Section 4.1), every interface matrix (16) is block-diagonal with two $n \times n$ blocks (the TE and TM polarizations). The Redheffer recursion and the cascade then run independently per polarization on $n \times n$ blocks — a further $4\times$ reduction on the (already reduced) patterned prefix and the assembly — and the result is reassembled as $\mathbf{S} = \text{blkdiag}(\mathbf{S}_{\text{TE}}, \mathbf{S}_{\text{TM}})$. Combined with the per-harmonic suffix of Section 3.2, the quasi-1D normal-incidence solve is thus scalar per harmonic throughout the uniform suffix and $n \times n$ -block in the prefix.

6 Binary gratings: low-rank permittivity structure

For the step-rect profile $\varepsilon(x) = \varepsilon_1$ on $|x| < x_1$, ε_2 otherwise, the Fourier coefficients follow from a one-line integral. Writing $\varepsilon(x) = \varepsilon_2 + \Delta\varepsilon \chi(x)$ with $\Delta\varepsilon = \varepsilon_1 - \varepsilon_2$ and χ the indicator of $|x| < x_1$,

$$\hat{\varepsilon}_m = \frac{1}{L_x} \int_{-L_x/2}^{L_x/2} \varepsilon(x) e^{-i2\pi mx/L_x} dx = \begin{cases} \varepsilon_2 + \Delta\varepsilon f, & m = 0, \\ \Delta\varepsilon f \text{sinc}(mf), & m \neq 0, \end{cases} \quad (48)$$

where $f = 2x_1/L_x$ is the fill fraction, $\text{sinc}(z) = \sin(\pi z)/(\pi z)$, and the second line uses $\int_{-x_1}^{x_1} e^{-i\beta x} dx = 2 \sin(\beta x_1)/\beta$ with $\beta = 2\pi m/L_x$. The Toeplitz convolution matrix is therefore $\mathbf{C} = \varepsilon_2 \mathbf{I} + \Delta\varepsilon f \mathbf{S}$ with the “discrete sinc” matrix $\mathbf{S}_{ij} = \text{sinc}(f(i-j))$.

The matrix $\mathbf{S}$ is the finite-dimensional prolate (Slepian) matrix [10]: its symbol is the indicator of the frequency band $|\theta| \leq \pi f$, and its eigenvalues obey the two-interval concentration law — approximately nf eigenvalues near 1 and the rest exponentially small,

$$\text{rank}_\tau(\mathbf{S}) \approx nf + O(\log \tau^{-1}). \quad (49)$$

For the EUV example of Section 9 ($f = 500/3500 \approx 0.143$, $n = 253$) this gives a numerical rank of ≈ 52 at $\tau = 10^{-14}$, i.e. its numerical rank is approximately five times smaller than its dimension; a rank-52 reconstruction reproduces $\mathbf{S}$ to $\sim 1.5 \times 10^{-15}$. Two consequences follow.

First, *the analytic coefficients* (48) replace the grid sampling and fast Fourier transform in the construction of C , and remove the pixelation error of a discrete grid (a 1000-pixel cell quantizes the fill to $f = 0.1420$ against the exact $f = 0.142857$). For smooth profiles (Gaussian, Lorentzian) the coefficients decay exponentially and the harmonic count required for a fixed accuracy drops correspondingly.

Second, *both patterned-layer eigenproblems admit diagonal-plus-low-rank representations*. Let $\mathbf{S} = U\Sigma U^\top$ be the (numerically) rank- r eigendecomposition, $U \in \mathbb{C}^{n \times r}$, $\Sigma \in \mathbb{R}^{r \times r}$. By the Woodbury identity

$$C^{-1} = (\varepsilon_2 \mathbf{I} + \Delta \varepsilon f U \Sigma U^\top)^{-1} = \frac{1}{\varepsilon_2} \left(\mathbf{I} - U (\varepsilon_2 \mathbf{I} + \Delta \varepsilon f \Sigma)^{-1} (\Delta \varepsilon f \Sigma) U^\top \right), \quad (50)$$

i.e. a diagonal plus a rank- r update, with the abbreviation $\Gamma = (\varepsilon_2 \mathbf{I} + \Delta \varepsilon f \Sigma)^{-1} \Delta \varepsilon f \Sigma$. Using $D = \text{diag}(\mathbf{k}_x)$,

$$C D C^{-1} D = D^2 + \frac{1}{\varepsilon_2} \left(\Delta \varepsilon f \mathbf{S} D^2 - \varepsilon_2 D U \Gamma U^\top D - \Delta \varepsilon f \mathbf{S} D U \Gamma U^\top D \right), \quad (51)$$

each bracketed term having rank at most r (a product of the rank- r factor $U \Gamma U^\top$ or $U \Sigma U^\top$ with diagonal matrices). Hence the patterned blocks (32) take the form

$$\mathbf{M}_{11} = \text{diag}(\beta_h) + \omega^2 \Delta \varepsilon f \mathbf{S}, \quad \mathbf{M}_{22} = \text{diag}(\beta'_h) + \text{rank-} \leq 4r \text{ update}, \quad (52)$$

with $\beta_h = \omega^2 \varepsilon_2 - \kappa_y^2 - k_{x,h}^2$. Equation (52) suggests matrix-vector products, subspace iteration, or secular-equation methods whose cost depends on the numerical rank r . However, the matrices in (32) are generally non-normal, and the existence of a low-rank update alone does not imply an $O(nr + r^3)$ algorithm for their complete spectrum. We therefore treat (52) as an operator-structure observation and do not include it in the exact speedup claims below. A stable reduced-rank eigensolver, especially for complex lossy media, is left for future work.

7 Block-wise post-processing

The absorbed power in a layer is $\text{val} = \text{tr}(\mathbf{ab} \mathbf{M} \mathbf{F}^\dagger \mathbf{M}_{\text{tot}} \mathbf{F})$, where $\mathbf{ab} = [\mathbf{a}_i; \mathbf{b}_i] \in \mathbb{C}^{4n}$ stacks the forward and backward modal amplitudes, $\mathbf{ab} \mathbf{M} = \mathbf{ab} \mathbf{ab}^\dagger \odot \mathbf{M}_t$ is a Hadamard (element-wise) product with the modal z -integral matrix $\mathbf{M}_t$, and $\mathbf{M}_{\text{tot}}$ is the block-diagonal material weighting. A direct evaluation materializes the $4n \times 4n$ matrices $\mathbf{M}_t$, $\mathbf{ab} \mathbf{M}$ and the $3n \times 3n$ matrices $\mathbf{M}_{\text{tot}}$, $\mathbf{F}$.

Using the block forms

$$\mathbf{F} = \begin{bmatrix} \mathbf{F}_{xy} & -\mathbf{F}_{xy} \\ \mathbf{F}_{az} & \mathbf{F}_{az} \end{bmatrix}, \quad \mathbf{M}_t = \begin{bmatrix} \mathbf{M}_{aa} & \mathbf{M}_{ab} \\ \mathbf{M}_{ab} & \mathbf{M}_{aa} \end{bmatrix}, \quad (53)$$

the interior product collapses onto $2n \times 2n$ blocks. Writing $\mathbf{T} = \mathbf{F}^\dagger \mathbf{M}_{\text{tot}} \mathbf{F}$ with $\mathbf{M}_{\text{tot}} = \text{blkdiag}(\mathbf{M}_{xy}, \mathbf{M}_z)$,

$$\mathbf{T} = \begin{bmatrix} \mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az} & -\mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az} \\ -\mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az} & \mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az} \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{bmatrix}, \quad (54)$$

with $\mathbf{A} = \mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az}$ and $\mathbf{B} = -\mathbf{F}_{xy}^\dagger \mathbf{M}_{xy} \mathbf{F}_{xy} + \mathbf{F}_{az}^\dagger \mathbf{M}_z \mathbf{F}_{az}$. The trace is then expanded without ever forming $\mathbf{ab} \mathbf{M}$:

$$\text{val} = \text{tr}((\mathbf{ab} \mathbf{ab}^\dagger \odot \mathbf{M}_t) \mathbf{T}) = \sum_{i,j} \text{ab}_i (\mathbf{M}_t \odot \mathbf{T}^\top)_{ij} \overline{\text{ab}_j} = \text{ab}^\top (\mathbf{M}_t \odot \mathbf{T}^\top) \overline{\text{ab}}, \quad (55)$$

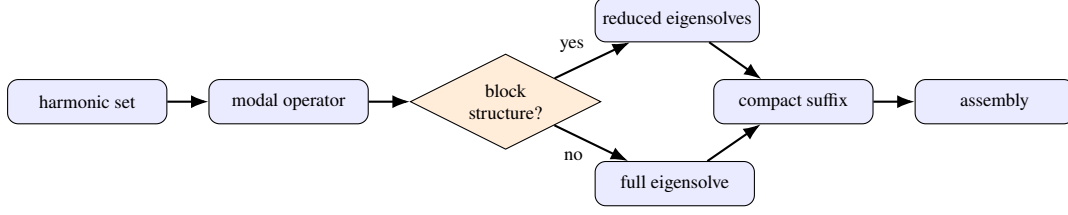


Figure 3: Structure-aware computational workflow. The branch decision is made from algebraic conditions on the discretized operator; if a condition is not met, the method falls back to the general formulation.

using $(\mathbf{X} \odot \mathbf{Y})_{ij} = \mathbf{X}_{ij} \mathbf{Y}_{ij}$ and $\text{tr}(\mathbf{Z}) = \sum_i \mathbf{Z}_{ii}$. Blockwise, with $\mathbf{W}_A = \mathbf{M}_{aa} \circ \mathbf{A}^\top$ and $\mathbf{W}_B = \mathbf{M}_{ab} \circ \mathbf{B}^\top$,

$$\text{val} = \mathbf{a}^\top \mathbf{W}_A \bar{\mathbf{a}} + \mathbf{b}^\top \mathbf{W}_A \bar{\mathbf{b}} + \mathbf{a}^\top \mathbf{W}_B \bar{\mathbf{b}} + \mathbf{b}^\top \mathbf{W}_B \bar{\mathbf{a}}. \quad (56)$$

The evaluation therefore needs two $2n \times 2n$ Hadamard products ($\mathbf{W}_A, \mathbf{W}_B$), the four block products that form $\mathbf{A}, \mathbf{B}$ in (54), and four matvecs — never materializing the $4n \times 4n$ and $3n \times 3n$ matrices. This is a $\sim 20\times$ reduction in transient memory for the absorption and stress-tensor integrals, at no change in the result (identical to the original formula to $\sim 10^{-15}$).

8 Implementation

The reductions above are implemented in a C++17 library using Eigen, FFTW, and LAPACK/BLAS. The remaining algorithmic considerations are:

8.1 Algorithmic workflow

The implementation follows the mathematical factorization rather than treating all layers uniformly. The workflow is:

1. construct the reciprocal lattice and select the retained harmonic set;
 2. form the convolution matrices only for patterned layers;
 3. classify each patterned modal operator by its measured off-diagonal block norms;
 4. solve the smallest valid eigenproblems and reconstruct the modal basis;
 5. locate the trailing homogeneous suffix and evaluate it by independent 2×2 recursions;
 6. compose the remaining patterned prefix with the compact suffix using the overlapping elimination in (21);
 7. reconstruct fields and quadratic observables from the resulting modal amplitudes.
- *Parallel eigenpairs.* The two independent $n \times n$ eigenproblems of Sections 4.1–4.2 are solved concurrently on two threads. OpenBLAS executes `zgeev` single-threaded (LAPACK-serial), so the pair runs $\sim 1.5\text{--}1.8\times$ faster with bit-identical results; with a threaded LAPACK (e.g. MKL) the parallelization is disabled by a flag.
 - *Star-product details.* The phase factors $\mathbf{d}_1, \mathbf{d}_2$ are kept as vectors and applied as diagonal scalings ($O(n)$ per product); matrix inversions in the star update are replaced by one *LU* factorization plus two back-substitutions; the products $\mathbf{d}_1 \mathbf{S}_{12}$ and $\mathbf{S}_{22} \mathbf{T}_{12}$ are hoisted as common subexpressions; and the LAPACK workspace is reused thread-locally.

Table 2: Validation against grcwa / S4.

case	R, T (this work)	reference	match
S4 golden T , p-pol	0.853173	0.852499*	$< 10^{-3}$
S4 golden T , s-pol	0.839654	0.839005*	$< 10^{-3}$
ex1 square holes	0.132415, 0.867585	0.13241493, 0.86758507	✓
ex2 two layers	0.132599, 0.867401	0.13259894, 0.86740106	✓
ex4 hexagonal	0.164450, 0.835550	0.16445004, 0.83554996	✓
EUV mirror 13.5 nm R	0.599171	0.599171	exact
EUV absorber R	0.578790	0.578790	$< 10^{-6}$

* S4 golden value embedded in the test; grcwa itself computes 0.853173 / 0.839654, reproduced exactly by this work.

- *Caching.* FFTW plans are cached per grid size; uniform layers with identical permittivity share their $(K, \mathbf{q}, \Phi)$; identical uniform–uniform interface matrices are reused; and the most recent S -matrix is memoized.
- *Memory.* Uniform layers deduplicate their operators, and the block-wise post-processing of Section 7 keeps the peak footprint bounded.

9 Results

9.1 Validation

Table 2 compares the solver with grcwa [9] and with values used by the S4 regression problem [3]. The grcwa examples are reproduced to the displayed precision. Fourier coefficients of the reflected field and fields reconstructed in real space differ by approximately 10^{-13} in the tested cases. The separate S4 values are satisfied within the 10^{-3} relative tolerance prescribed by that regression test.

The EUV absorber convergence with the harmonic count is $R = 0.5668, 0.5788, 0.5791, 0.57925$ for $n = 9, 97, 197, 301$ respectively — the per-harmonic suffix keeps the solve cost $O(n)$ in the number of uniform layers at every truncation.

9.2 Performance

Table 3 reports the effect of the per-harmonic uniform suffix (Section 3.2) on the full-2D EUV absorber (vacuum | TaN absorber | Ru cap | $40 \times (\text{Mo/Si})$ | Si substrate). The suffix alone accelerates the S -matrix assembly by 3.4–3.7 $\times$ with bit-identical reflectivity; the gain grows with n because the suffix cost is now linear in n while the reference dense products scale as $(2n)^3$.

The benchmark geometry uses wavelength $\lambda = 13.5$ nm and the layer sequence vacuum | TaN absorber | Ru cap | 40 periods of Mo/Si | Si substrate. Timings correspond to release-mode C++ code linked to OpenBLAS and use six BLAS threads. Each table entry is the best measured wall time over repeated runs on the same workstation; it should therefore be interpreted as an implementation benchmark rather than a hardware-independent constant. The harmonic count n is the actual count after symmetry-preserving truncation, and both columns use the same n , material grids, incidence, and floating-point precision.

On the quasi-one-dimensional geometry (the same absorber as a y -invariant bar, $L_x = 3.5 \mu\text{m}$, $n = 113$), the combined one-dimensional harmonic selection, per-harmonic suffix, and polarization decomposition reduce the end-to-end time by approximately a factor of 30 relative to grcwa for the tested configuration, at the same reflectivity $R = 0.528927$. This comparison includes implementation-language and library effects in addition to the algebraic reductions; the isolated full-two-dimensional suffix speedups are those reported in Table 3.

Table 3: Full-2D EUV absorber, S -matrix (`rt_solve`) before/after the per-harmonic uniform suffix (machine under load; R identical in both columns).

n	generic	per-harmonic suffix	speedup
201	1354 ms	369 ms	3.7×
301	3268 ms	964 ms	3.4×

10 Discussion

The per-harmonic uniform-suffix factorization and the exact eigenproblem decompositions are exact: they change only the *order* of arithmetic, not the model, and the scattering observables are bit-identical to the generic implementation. Three caveats bound their applicability.

First, the block-lower-triangular decomposition (Section 4.2) requires the harmonic set to be the one-dimensional row of a y -invariant grating, for which $\mathbf{k}_y$ is constant. For a genuinely two-dimensional pattern at oblique incidence the upper-right block is

$$\mathbf{M}_{12} = C \operatorname{diag}(\mathbf{k}_y) E \operatorname{diag}(\mathbf{k}_x) - \operatorname{diag}(\mathbf{k}_x \mathbf{k}_y), \quad (57)$$

which does not generally vanish. The implementation then uses the full $2n \times 2n$ problem; the choice is made from the norm of the computed block.

Second, the block-diagonal decomposition applies to the one-dimensional normal- incidence case and to additional patterns only when the computed off-diagonal blocks vanish. For a general two-dimensional pattern neither spectral reduction applies, and the full $2n \times 2n$ `zgeev` remains the recognized residual bottleneck. A matrix-free Krylov formulation (preconditioned by the quasi-1D solves above) is the natural next step for very large n , but is beyond the scope of this paper.

Third, the low-rank representation of Section 6 has not been used to replace the dense non-Hermitian eigensolver. Doing so requires a stability and conditioning analysis, particularly for lossy materials and near-degenerate modes. It should therefore be regarded as a direction for a reduced-order method rather than as part of the present exact algorithm.

Finally, we note that Li’s factorization rules [7] and the normal-vector method [8] improve *convergence in n* for high-contrast metallic gratings (reducing the required n), whereas the present reductions reduce the *cost per harmonic*. The two are orthogonal and compound: every harmonic saved by better factorization is processed at the reduced per-harmonic cost.

11 Conclusion

We have identified invariant subspaces of the finite-dimensional Maxwell operator that remove avoidable dense algebra from Fourier-modal scattering. Homogeneous layers form a direct sum over diffraction harmonics, and a y -invariant patterned layer has an exactly reducible spectrum at both normal and conical incidence. These identities preserve the Fourier discretization and its physical observables. In the tested EUV stack, the homogeneous-suffix algorithm reduces scattering time by 3.4–3.7 for full two-dimensional truncations; additional one-dimensional reductions produce substantially larger gains for line gratings. The remaining asymptotic cost is the dense non-Hermitian modal problem of a genuinely two-dimensional patterned layer and the final coupling of that layer to its exterior channels. Matrix-free boundary formulations and structure-preserving iterative eigensolvers are the natural next steps.

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